



A HYBRID AI-BASED PREDICTIVE MODEL FOR EARLY CARDIOVASCULAR DISEASE PREDICTION USING IOT SENSORS

A PROJECT REPORT

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ABSTRACT

Cardiovascular diseases (CVDs) are among the leading causes of death globally, posing a significant challenge to modern healthcare systems. One of the major difficulties in managing cardiovascular conditions is the lack of early detection, as symptoms often remain unnoticed until the disease reaches a critical stage. Traditional diagnostic approaches rely heavily on periodic hospital visits, clinical tests such as electrocardiograms (ECG), blood pressure measurements, and laboratory investigations. While these methods are effective for diagnosis, they do not support continuous monitoring of a patient's health, leading to missed opportunities for early intervention and prevention.

To address these limitations, this project proposes a hybrid artificial intelligence (AI)-based predictive model integrated with Internet of Things (IoT) sensors for early cardiovascular disease prediction and continuous health monitoring. The system is designed to collect real-time physiological data using multiple sensors, including ECG sensors for heart activity, blood pressure sensors for measuring systolic and diastolic values, pulse oximeters for oxygen saturation (SpO_2), and accelerometers for tracking physical activity. These sensors work together to provide a comprehensive understanding of an individual's cardiovascular health.

The collected data is transmitted through IoT communication technologies such as Wi-Fi or Bluetooth to a backend system developed using Python. In the backend, the raw sensor data undergoes preprocessing steps such as noise filtering, data cleaning, normalization, and feature extraction to ensure accuracy and reliability. The processed data is then analyzed using a hybrid machine learning model that combines multiple algorithms to improve prediction accuracy. This model identifies patterns and risk factors associated with cardiovascular diseases and classifies individuals based on their risk levels.

சுருக்கம்

இதய மற்றும் இரத்த நாள நோய்கள் (Cardiovascular Diseases - CVDs) உலகளவில் அதிக மரணங்களுக்கு காரணமாக இருக்கும் முக்கியமான சுகாதார பிரச்சினைகளில் ஒன்றாகும். இந்நோய்களை ஆரம்ப நிலையிலேயே கண்டறிவது மிகவும் சிரமமானதாக உள்ளது, ஏனெனில் பல நேரங்களில் அறிகுறிகள் தெளிவாக தெரியாமல் போகின்றன. பாரம்பரிய மருத்துவ முறைகள் பொதுவாக மருத்துவமனைக்கு நேரடியாக சென்று பரிசோதனை செய்வதை மட்டுமே சார்ந்துள்ளன. ECG, இரத்த அழுத்தம் (Blood Pressure) மற்றும் ஆய்வக பரிசோதனைகள் போன்றவை பயனுள்ளதாக இருந்தாலும், அவை தொடர்ச்சியான கண்காணிப்பை வழங்குவதில்லை. இதனால் ஆரம்ப நிலையில் நோயை கண்டறியும் வாய்ப்புகள் குறைகின்றன.

இந்த பிரச்சினைகளை தீர்க்க, இத்திட்டம்செயற்கை நுண்ணறிவு (AI) மற்றும் பொருட்களின் இணையம் (IoT) இணைந்த ஹைபிரிட் முன்னறிவிப்பு முறைமையைமுன்வைக்கிறது. இந்த முறைமையின் மூலம் நோயாளியின் உடல் நிலையை தொடர்ச்சியாக கண்காணித்து இதய நோய்களின் அபாயத்தை ஆரம்பத்திலேயே கண்டறிய முடியும். ECG சென்சார், இரத்த அழுத்த சென்சார், SpO₂ (ஆக்சிஜன் அளவு) சென்சார் மற்றும் ஆக்சிலரோமீட்டர் போன்ற பல்வேறு சென்சார்கள் பயன்படுத்தி உடலின் முக்கிய தரவுகள் நேரடியாக சேகரிக்கப்படுகின்றன. இந்த தரவுகள் ஒருங்கிணைந்து ஒருவரின் இதய ஆரோக்கியத்தை முழுமையாக புரிந்து கொள்ள உதவுகின்றன.

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LIST OF ABBREVIATIONS

ECG	ELECTROENCEPHALOGRAPHY
CNN	CONVOLUTION NEURAL NETWORK
AI	ARTIFICIAL INTELLIGENCE
ML	MACHINE LEARNING

CHAPTER1

INTRODUCTION

Cardiovascular diseases (CVDs) are one of the leading causes of death worldwide, affecting millions of people every year. These diseases include conditions such as heart attacks, hypertension, and other disorders related to the heart and blood vessels. One of the major challenges in dealing with cardiovascular diseases is that their early symptoms are often mild or unnoticed, which leads to late diagnosis and increases the risk of severe complications. Early detection and continuous monitoring of heart health are therefore essential to reduce mortality rates and improve patient outcomes.

Traditional healthcare systems mainly rely on periodic medical checkups and hospital-based diagnostic methods such as electrocardiograms (ECG), blood pressure measurements, and laboratory tests. While these techniques are effective, they are not designed for continuous monitoring in daily life.

Patients are usually examined only when they visit a hospital, and there is no proper system to track their health condition in real time. This gap makes it difficult to identify early warning signs of cardiovascular diseases and delays timely medical intervention. With the rapid advancement of technology, the integration of the Internet of Things (IoT) and Artificial Intelligence (AI) has opened new possibilities in the healthcare sector. IoT enables the use of smart sensors to collect real-time physiological data from patients, while AI techniques, particularly machine learning, can analyze large amounts of data to identify patterns and predict potential health risks. These technologies together provide an opportunity to develop intelligent healthcare systems that can monitor, analyze, and predict diseases effectively.

This project focuses on developing a hybrid AI-based predictive model integrated with IoT sensors for early cardiovascular disease prediction. The system uses various sensors such as ECG, blood pressure, pulse oximeter

(SpO₂), and accelerometer to continuously monitor vital health parameters. The collected data is transmitted to a backend system where it is processed and analyzed using machine learning algorithms to predict the risk level of cardiovascular disease.

In addition, the system includes a user-friendly interface developed using Streamlit, which allows real-time visualization of health data and prediction results. This enables both patients and healthcare professionals to monitor health conditions remotely and take necessary actions when abnormalities are detected. The system also supports alert mechanisms to notify users in case of critical health conditions.

Overall, this project aims to provide a cost-effective, portable, and efficient solution for continuous health monitoring and early disease prediction. By combining IoT and AI technologies, the proposed system enhances preventive healthcare, reduces dependency on frequent hospital visits, and contributes to improving the overall quality of life.

CHAPTER 2

LITERATURE SURVEY

A literature survey is an important part of research that provides an overview of existing studies related to a particular topic. It helps in understanding current technologies, methodologies, advantages, limitations, and research gaps in the field of cardiovascular disease prediction using Artificial Intelligence (AI), Machine Learning (ML), and Internet of Things (IoT) technologies. The following research papers were studied to analyze various approaches used for cardiovascular disease prediction and healthcare monitoring systems.

1. An RNN-BiLSTM Based Multi-Decision GAN Approach for the Recognition of Cardiovascular Disease from Heartbeat Sound (2024)

Authors : Vinay, N. A., et al.

Publisher : IEEE Access

This research paper proposes an advanced deep learning-based approach for recognizing cardiovascular diseases using heartbeat sound signals. The authors developed a hybrid model integrating Recurrent Neural Networks (RNN), Bidirectional Long Short-Term Memory (BiLSTM), and Generative Adversarial Networks (GAN) for accurate cardiovascular disease detection.

The proposed system analyzes heartbeat sound recordings collected from patients and extracts important audio features related to cardiac abnormalities. The BiLSTM model helps in capturing both forward and backward temporal dependencies in heartbeat signals, improving classification performance. GAN techniques are additionally used to improve data quality and generate synthetic training samples for enhancing model accuracy.

The study highlights that deep learning techniques can effectively identify cardiovascular disorders from heartbeat sounds with improved accuracy compared to conventional machine learning approaches. The model achieved better performance.

However, the study mainly focuses on heartbeat sound analysis and requires high-quality audio datasets for accurate prediction. The research strongly supports the use of deep learning techniques in intelligent healthcare systems for early cardiovascular disease diagnosis.

performance of multiple supervised learning techniques including Decision Trees, Random Forest, **2. Latent Space Classification for Cardiovascular Disease Detection Using Deep Convolutional Autoencoders (2025)**

Authors : A. G. Gracy and S. C. Pravin

Publisher : IEEE Access

This paper presents a deep learning framework for cardiovascular disease detection using Deep Convolutional Autoencoders and latent space classification techniques. The proposed system focuses on extracting hidden and meaningful representations from healthcare datasets to improve disease prediction performance.

The authors utilize convolutional autoencoders for automatic feature extraction and dimensionality reduction. The extracted latent features are further classified using machine learning classifiers to identify cardiovascular disease conditions. The study demonstrates that deep feature extraction methods significantly improve prediction accuracy and reduce computational complexity.

The paper emphasizes the importance of efficient feature learning in medical diagnosis systems. The proposed approach achieved high classification accuracy while handling complex healthcare datasets containing multiple physiological parameters.

The study also discusses the advantages of deep learning in reducing manual feature engineering efforts. However, the system requires large-scale training data and computational resources for optimal performance. This research contributes to the development of intelligent AI-driven cardiovascular prediction systems.

3. Effective Heart Disease Prediction Using Machine Learning Techniques (2023)

Authors : C. M. Bhatt, et al.

Publisher : Algorithms

This research paper focuses on predicting heart disease using various machine learning algorithms and healthcare datasets. The study compares the Support Vector Machine (SVM), Logistic Regression, and Naïve Bayes algorithms.

The proposed system analyzes patient medical data such as age, cholesterol levels, blood pressure, heart rate, and other clinical parameters to predict the possibility of heart disease occurrence. The authors performed preprocessing operations including data cleaning, normalization, and feature selection to improve prediction performance.

The study found that ensemble learning methods such as Random Forest achieved better prediction accuracy compared to traditional classification algorithms. The paper also highlights the importance of feature selection and preprocessing in improving machine learning model performance.

One of the major advantages of this research is its focus on practical healthcare datasets and interpretable machine learning models. However, the study identifies challenges related to missing values and dataset imbalance. The research demonstrates that machine learning techniques can effectively support early cardiovascular disease prediction and preventive healthcare.

4. Early Heart Disease Prediction Using Feature Engineering and Machine Learning Algorithms (2024)

Authors : M. A. Bouqentar, et al.

Publisher : Heliyon

This paper presents a machine learning-based framework for early heart disease prediction using feature engineering techniques and predictive analytics. The

research emphasizes the role of feature extraction and feature optimization in improving the performance of cardiovascular disease prediction models.

The authors applied several machine learning algorithms including Random Forest, Support Vector Machine (SVM), Logistic Regression, and Decision Tree classifiers to analyze patient healthcare data. Feature engineering techniques were used to identify the most relevant clinical attributes affecting cardiovascular disease prediction.

The study demonstrates that optimized feature selection significantly improves model accuracy and reduces prediction errors. The proposed framework achieved reliable prediction performance for identifying high-risk cardiovascular patients at an early stage.

The paper also discusses the importance of preprocessing techniques such as normalization and missing value handling in healthcare analytics. However, the research suggests that future systems should integrate real-time IoT sensor data for continuous monitoring applications. The study strongly supports the integration of AI and feature engineering techniques for intelligent healthcare systems.

5. Machine Learning Prediction in Cardiovascular Diseases: A Meta-Analysis (2020)

Authors : C. Krittanawong, et al.

Publisher : Scientific Reports

This research paper presents a comprehensive meta-analysis of machine learning applications in cardiovascular disease prediction and diagnosis. The study reviews multiple research works related to AI-based cardiovascular disease detection and evaluates the effectiveness of different machine learning algorithms in healthcare applications.

The paper analyzes machine learning techniques such as Neural Networks, Random Forest, Support Vector Machine (SVM), Gradient Boosting, and Deep

Learning models used in cardiovascular healthcare systems. The study highlights that machine learning algorithms can provide high prediction accuracy and support early disease diagnosis.

The authors emphasize that AI-driven healthcare systems can improve clinical decision-making, reduce mortality rates, and support personalized healthcare management. The research also discusses the growing importance of wearable devices, IoT-based monitoring systems, and real-time healthcare analytics in modern medical applications.

However, the study identifies several challenges including lack of standardized medical datasets, data privacy concerns, model interpretability issues, and computational complexity. The paper concludes that integrating AI, IoT, and real-time monitoring technologies can significantly improve cardiovascular disease prediction and preventive healthcare systems.

CHAPTER 3

SYSTEM ANALYSIS

The system analysis of the proposed hybrid AI-based cardiovascular disease prediction model focuses on understanding the requirements, limitations, and functional aspects of integrating IoT sensors with machine learning techniques for real-time health monitoring. The existing system relies on periodic hospital visits and lacks continuous monitoring, leading to delayed detection of heart-related issues. To overcome these limitations, the proposed system analyzes the need for continuous data collection through sensors such as ECG, blood pressure, SpO₂, and accelerometer, along with efficient data transmission using IoT technologies. It also evaluates the processing requirements for handling real-time data, including preprocessing, feature extraction, and prediction using a hybrid machine learning model. The system ensures user-friendly interaction through a Streamlit dashboard for visualization and includes alert mechanisms for abnormal conditions. Additionally, considerations such as data accuracy, reliability, security, scalability, and cost-effectiveness are analyzed to ensure the system performs efficiently in real-world healthcare scenarios.

3.1 EXISTING SYSTEM

The existing system for cardiovascular disease detection and monitoring is primarily based

on traditional healthcare practices that rely on hospital-based examinations and periodic medical checkups. In this approach, patients are required to visit healthcare facilities to undergo diagnostic tests such as electrocardiograms (ECG), blood pressure measurements, cholesterol level analysis, blood sugar tests, and imaging techniques like echocardiography or angiography.

These tests are conducted using specialized medical equipment and require the presence of trained healthcare professionals for accurate diagnosis and interpretation of results. While these methods are considered reliable and

clinically validated, they are mainly focused on diagnosing diseases after symptoms have already appeared rather than preventing them at an early stage.

One of the major limitations of the existing system is the lack of continuous monitoring. Patients are typically examined only during scheduled appointments or when they experience noticeable symptoms. This creates a gap between checkups, during which important physiological changes or early warning signs may go undetected. As cardiovascular diseases often develop silently over time, the absence of real-time monitoring increases the risk of late diagnosis, which can lead to severe complications such as heart attacks or strokes. Additionally, many individuals may ignore minor symptoms or delay hospital visits due to lack of awareness, financial constraints, or limited access to healthcare services.

Another drawback of the existing system is its dependency on hospital infrastructure and medical resources. Diagnostic procedures can be time-consuming and costly, making it difficult for patients to undergo frequent checkups. This is particularly challenging for people living in rural or remote areas where advanced medical facilities and specialists are not easily available.

The current approach also lacks automation and intelligent analysis. Most diagnoses depend on manual interpretation by doctors, which may vary based on experience and workload. There is minimal use of advanced technologies such as artificial intelligence or machine learning for predicting disease risk based on historical and real-time data. As a result, the existing system is reactive rather than proactive, focusing more on treatment than prevention.

In summary, although the existing system provides accurate diagnostic results, it suffers from several limitations, including lack of continuous monitoring, delayed detection, high cost, limited accessibility, and absence of predictive

3.2 PROBLEM STATEMENT

Cardiovascular diseases (CVDs) are one of the leading causes of death worldwide, and their early detection remains a major challenge in the healthcare sector. In many cases, the initial symptoms of heart-related problems are mild, silent, or go unnoticed, resulting in late diagnosis and increased risk of severe complications such as heart attacks or strokes. The existing healthcare system mainly depends on periodic hospital visits and clinical examinations, which do not provide continuous monitoring of a patient's health condition. This lack of real-time monitoring makes it difficult to identify early warning signs and delays timely medical intervention.

Additionally, traditional diagnostic methods require specialized medical equipment and trained professionals, making them expensive and less accessible, especially for people living in rural or remote areas. Patients often face difficulties in undergoing regular health checkups due to time constraints, financial limitations, and limited availability of healthcare facilities. As a result, many individuals remain unaware of their cardiovascular health status until the condition becomes critical.

Another significant issue is the absence of intelligent and automated systems for predicting cardiovascular disease risk. Current systems focus more on diagnosis rather than prevention and do not effectively utilize modern technologies such as Artificial Intelligence (AI) and the Internet of Things (IoT) for continuous health monitoring and predictive analysis. Furthermore, there is a lack of integrated systems that can collect, process, and analyze real-time physiological data such as ECG, blood pressure, oxygen levels, and physical activity.

Therefore, there is a need for a cost-effective, portable, and intelligent system that can continuously monitor vital health parameters, provide real-time

analysis, and predict the risk of cardiovascular diseases at an early stage. Such a system would enable timely medical intervention, improve patient outcomes, and support preventive healthcare by reducing the dependency on traditional hospital-based diagnosis methods.

3.3PROPOSED SYSTEM

The proposed system is a hybrid Artificial Intelligence (AI) and Internet of Things (IoT)-based solution designed for early prediction and continuous monitoring of cardiovascular diseases. This system aims to overcome the limitations of traditional healthcare methods by enabling real-time data collection, intelligent analysis, and remote health monitoring. It integrates multiple physiological sensors such as Electrocardiogram (ECG), blood pressure sensor, pulse oximeter (SpO₂), and accelerometer to continuously track vital health parameters of the user.

The sensors are connected to a microcontroller unit (such as NodeMCU), which collects and transmits the physiological data through IoT communication technologies like Wi-Fi or Bluetooth. The collected data is sent to a backend system developed using Python, where it undergoes preprocessing steps including noise removal, data cleaning, normalization, and feature extraction. These steps ensure that the data is accurate and suitable for further analysis.

A hybrid machine learning model is then applied to the processed data to analyze patterns and predict the risk level of cardiovascular diseases. This model combines multiple algorithms to improve prediction accuracy and reliability. Based on the analysis, the system classifies the user's health condition into different risk categories, enabling early detection of potential heart-related issues.

To enhance user interaction, the system includes a Streamlit-based dashboard that provides real-time visualization of health parameters such as heart rate, blood pressure, oxygen levels, and activity data. This interface allows both patients and healthcare professionals to monitor health conditions remotely and make informed decisions. Additionally, the system incorporates an alert and notification mechanism that generates warnings when abnormal values are detected, ensuring timely medical attention.

The proposed system is designed to be cost-effective, portable, and user-friendly, making it suitable for daily use and accessible to a wide range of users, including those in rural and remote areas. healthcare, reduces the need for frequent hospital visits, and improves the overall efficiency of cardiovascular disease management.

CHAPTER 4

REQUIREMENT SPECIFICATIONS

4.1 INTRODUCTION

Requirement analysis determines the requirements of a new system. This project analyses product and resource requirements, which is required for this successful system. The product requirement includes input and output requirements it gives the wants in terms of input to produce the required output. The resource requirements give in brief about the software and hardware that are needed to achieve the required functionality.

HARDWARE AND SOFTWARE SPECIFICATION

HARDWARE REQUIREMENTS

The proposed system requires both computing components and biomedical sensing devices to collect and process real-time physiological data. The hardware components are selected to ensure accurate monitoring, efficient data transmission, and system reliability.

NodeMCU (ESP8266)	Used as the main microcontroller for data collection and IoT
ECG Sensor	Measures the electrical activity of the heart
Blood Pressure Sensor	Monitors systolic and diastolic blood pressure
Pulse Oximeter (SpO₂ Sensor)	Measures oxygen saturation and pulse rate
Accelerometer (MEMS Sensor)	Tracks physical activity and movement
LCD Display	Displays real-time sensor readings
Buzzer	Provides alert notifications during abnormal conditions
Power Supply Unit	Supplies required power to all components
Connecting Wires & Breadboard	For circuit connections

Table 4.1: Hardware Requirements

SOFTWARE REQUIREMENTS

The software components are responsible for data processing, machine learning analysis, and visualization of health parameters. These tools ensure smooth functioning of the system from data acquisition to prediction and monitoring.

Programming Language	Python
Development Environment	Jupyter Notebook / VS Code / PyCharm
Framework	Streamlit (for dashboard visualization)
Machine Learning Libraries	Scikit-learn, TensorFlow / Keras
Data Processing Libraries	NumPy, Pandas
Visualization Tools	Matplotlib, Seaborn
IoT Communication	MQTT / HTTP Protocols
Power Supply Unit	Supplies required power to all components
Operating System	Windows / Linux

Table 4.2: Software Requirements

TECHNOLOGIES USED : Python, OpenCV, and Mediapipe PYTHON

Python is an interpreted language, which means that it does not require compilation, making it quick and easy to test and debug code. It has a vast and active community of developers who contribute to its extensive library of modules and packages, making it versatile and suitable for a wide range of applications. Python is also used in data science, machine learning, and artificial intelligence, making it a valuable tool for developers in these fields.

INTRODUCTION TO PYTHON

Python is a high-level programming language that is widely used in various fields, including web development, data analysis, scientific computing, and artificial intelligence. It was first released in 1991 by Guido van Rossum, and it has since become one of the most popular programming languages in the world.

One of the reasons why Python is so popular is its simplicity and ease of use. Python has a simple and intuitive syntax that makes it easy to learn and write code. Its syntax is designed to be readable and expressive, which means that code written in Python is often easier to understand and maintain than code written in other languages.

Python is also an interpreted language, which means that it does not need to be compiled before it can be run. This makes it quick and easy to test and debug code. Python's interpreter reads and executes the code line-by-line, which means that developers can see the output of their code immediately.

Python is a versatile language that can be used for a wide range of applications. It has a large and active community of developers who contribute to its extensive library of modules and packages. These modules and packages can be used for various purposes and regular expressions.

Python is an object-oriented language, which means that it allows for the creation of reusable and modular code. Object-oriented programming (OOP) is a programming paradigm that emphasises the use of objects, which are instances of classes, to represent real-world entities and their interactions. Python's support for OOP allows developers to write code that is easy to read, maintain, and extend.

Python is also a cross-platform language, which means that it can run on different operating systems, such as Windows, Mac OS, and Linux, without requiring any modifications to the code. This makes it easy for developers to write code that can run on different platforms and devices.

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In conclusion, Python is a versatile, easy-to-learn, and widely used programming language that has a large and active community of developers. Its simplicity, versatility, and cross-platform compatibility make it a popular choice for a wide range of applications, from web development to artificial intelligence.

DESIGN PHILOSOPHY AND FEATURES

Python is a multi-paradigm programming language. Object-oriented programming and structured programming are fully supported, and many of their features support functional programming and aspect-oriented programming (including metaprogramming and metaobjects). Many other paradigms are supported via extensions, including design by contract and logic programming.

Python uses dynamic typing and a combination of reference counting and a cycle-detecting garbage collector for memory management. It uses Python is also a cross-platform language, which means that it can run on different operating systems, such as Windows, Mac OS, and Linux, without requiring any modifications to the code. This makes it easy for developers to write code that can run on different platforms and devices.

dynamic name resolution (late binding), which binds method and variable names during program execution.

Its design offers some support for functional programming in the Lisp tradition. It has filter, map and reduce functions; list comprehensions, dictionaries, sets, and generator expressions. The standard library has two modules (itertools and functools) that implement functional tools borrowed from Haskell and Standard ML.

Its core philosophy is summarized in the document The Zen of Python (PEP 20), which includes aphorisms such as:

- Beautiful is better than ugly.
- Explicit is better than implicit.
- Simple is better than complex.
- Complex is better than complicated.
- Readability counts.

Rather than building all of its functionality into its core, Python was designed to be highly extensible via modules. This compact modularity has made it particularly popular as a means of adding programmable interfaces to existing applications. Van Rossum's vision of a small core language with a large standard library and easily extensible interpreter stemmed from his frustrations with ABC, which espoused the opposite approach.

Python strives for a simpler, less-cluttered syntax and grammar while giving developers a choice in their coding methodology. In contrast to Perl's "there is more than one way to do it" motto, Python embraces a "there should be one—and preferably only one—obvious way to do it" philosophy.

Python's developers strive to avoid premature optimization and reject patches to non-critical parts of the C, Python reference implementation that would offer marginal increases in speed at the cost of clarity. When speed is important, a Python programmer can move time-critical functions to extension modules written in languages a just-in-time compiler. Python is also available, which translates a Python script into C and makes direct C-level API calls into the Python interpreter.

Python's developers aim for it to be fun to use. This is reflected in its name—a tribute to the British comedy group Monty Python —and in occasionally playful approaches to tutorials and reference materials, such as the use of the terms "spam" and "eggs" (a reference to a Monty Python sketch) in examples, instead of the often-used "foo" and "bar".

A common neologism in the Python community is *pythonic*, which has a wide range of meanings related to program style. "Pythonic" code may use Python idioms well, be natural or show fluency in the language, or conform with Python's minimalist philosophy and emphasis on readability. Code that is difficult to understand or reads like a rough transcription from another programming language is called *unpythonic*.

LIBRARIES

Python's large standard library provides tools suited to many tasks and is commonly cited as one of its greatest strengths. For Internet-facing applications, many standard formats and protocols such as MIME and HTTP are supported. It includes modules for creating graphical user interfaces, connecting to relational databases, generating pseudo random numbers, arithmetic with arbitrary-precision decimals, manipulating regular expressions, and unit testing.

Some parts of the standard library are covered by specifications—for example, the Web Server Gateway Interface (WSGI) implementation `wsgiref` follows PEP 333—but most are specified by their code, internal documentation, and test suites.

Some parts of the standard library are covered by specifications—for example, the Web Server Gateway Interface (WSGI) implementation `wsgiref` follows PEP 333—but most are specified by their code, internal documentation, and test suites. However, because most of the standard library is cross-platform Python code, only a few modules need altering or rewriting for variant implementations.

As of 14 November 2022, the Python Package Index (PyPI), the official repository for third-party Python software, contains over 415,000 packages with a wide range of functionality, including:

- Automation
- Data analytics
- Data bases
- Documentation
- Graphical user interfaces
- Image processing
- Machine learning
- Mobile apps
- Multimedia
- Computer networking
- Scientific computing
- System administration
- Test frameworks
- Text processing

- Web frameworks
- Web scraping

WORKING OF PYTHON

Python is a high-level programming language that is designed to be easy to read and write. It is an interpreted language, which means that it does not need to be compiled before it can be run. Instead, Python's interpreter reads and executes the code line-by-line.

When a developer writes a program in Python, they write the code in a text editor and save it as a Python file with a .py extension. They can run the program by opening a terminal or command prompt and typing "python" followed by the name of the file.

When the program is run, Python's interpreter reads the code line-by-line and executes it. Python uses indentation to group statements together, which means that the code is easy to read and understand.

Python also has a largest and a rd library and a vast collection of third-party modules and packages. These modules and packages provide additional functionality that can be easily integrated into a Python program.

Python's versatility and ease of use make it a popular choice for a wide range of applications. It is used for web development, data analysis, scientific computing, machine learning, and artificial intelligence, among others. OpenCV's object detection and recognition capabilities are also widely used. It provides a variety of functions and algorithms for object detection and recognition, such as Haar cascades, which are widely used for face detection.

INTRODUCTION TO OPENCV

OpenCV (Open Source Computer Vision Library) is an open-source computer vision and machine learning software library. It was first released in 2000 by Intel, and since then, it has become one of the most popular libraries for computer vision tasks.

OpenCV is written in C++, but it has bindings for Python, Java, and MATLAB, among other languages. It provides a wide range of functions and algorithms that can be used for various computer vision tasks, such as image and video processing, object detection and recognition, facial recognition, and motion tracking.

OpenCV can be used in a wide range of applications, from robotics and surveillance to medical imaging and augmented reality. It has a large and active community of developers who contribute to its development and maintenance, making it a versatile and reliable tool for computer vision tasks.

OpenCV provides a variety of functions and algorithms that can be used for image and video processing, such as image filtering, thresholding, and edge detection. It also provides functions for feature detection and extraction, such as SIFT (Scale-Invariant Feature Transform) and SURF (Speeded Up Robust Features).

OpenCV's object detection and recognition capabilities are also widely used. It provides a variety of functions and algorithms for object detection and recognition, such as Haar cascades, which are widely used for face detection.

OpenCV's machine learning capabilities are also becoming increasingly popular. It provides a variety of functions and algorithms for machine learning, such as decision trees, support vector machines, and deep neural networks.

Overall, OpenCV is a versatile and widely used library for computer vision and machine learning tasks. Its large and active community of developers, extensive documentation, and ease of use make it a valuable tool for developers and researchers in various fields.

APPLICATIONS

OpenCV's application are as include:

- 2D and 3D feature tool kits
- Ego motion estimation
- Facial recognition system
- Gesture recognition
- Human–computer interaction (HCI)
- TensorFlow is an open-source machine learning and deep learning framework
- Mobile robotics
- Motion understanding
- Object detection
- Segmentation and recognition
- Stereopsisstereovision : depthperception from 2 cameras
- Structure from motion (SFM)
- Motion video tracking
- Augmented reality

To support some of the above areas, OpenCV includes a statistical machine learning library that contains:

- Boosting

- Decision tree learning
- Gradient boosting trees
- Expectation-maximization algorithm
- k-nearest neighbor algorithm
- Naïve Bayes classifier
- Artificial neural networks
- Random forest
- Support vector machine (SVM)
- Deep neural networks (DNN)

TensorFlow

developed by the Google Brain team. It is widely used for building, training, and deploying machine learning models, especially in applications involving large-scale data and complex computations. TensorFlow provides a flexible architecture that allows developers to run computations on multiple platforms such as CPUs, GPUs, and even mobile devices. It is particularly popular in healthcare, image processing, natural language processing, and predictive analytics due to its scalability and efficiency.

Description

TensorFlow works based on the concept of data flow graphs, where mathematical operations are represented as nodes and the data (tensors) flows between them. A tensor is a multi-dimensional array that represents data, such as images, signals, or numerical values. In the context of this project, TensorFlow is used to process physiological data collected from IoT sensors and to build machine learning models for predicting cardiovascular disease risk.

building, training, and evaluation. It supports deep learning models such as

Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). TensorFlow also includes features like automatic differentiation, which simplifies the optimization of models during training.

One of the major advantages of TensorFlow is its ability to handle large datasets efficiently and perform parallel computations, making it suitable for real-time health monitoring systems. It also provides TensorFlow Lite, which allows models to be deployed on embedded systems and mobile devices, making it ideal for IoT-based healthcare applications.

In this project, TensorFlow plays a key role in analyzing sensor data, identifying patterns, and predicting the risk of cardiovascular diseases with high accuracy. Its scalability and performance make it an essential component in building an intelligent healthcare monitoring system.

Keras

Introduction

Keras is a high-level deep learning API that runs on top of TensorFlow and simplifies the process of building and training neural networks. It was developed by François Chollet and is designed to be user-friendly, modular, and easy to understand. Keras enables developers to quickly create deep learning models with minimal code, making it highly suitable for beginners as well as professionals.

Description

Keras provides a simple and intuitive interface for designing neural network architectures. It allows users to build models using a sequential approach (layer-by-layer) or a functional approach for more complex architectures. In this project, Keras is used to design and implement the hybrid machine learning model that predicts cardiovascular disease risk based on sensor data.

The library includes a wide range of pre-built layers such as dense layers, dropout layers, convolutional layers, and activation functions. It also provides built-in support for loss functions, optimizers, and evaluation metrics, which simplifies the model training process. Keras automatically handles many low-level operations, allowing developers to focus on model design and experimentation.

Another important feature of Keras is its seamless integration with TensorFlow, which allows it to leverage TensorFlow's powerful backend for efficient computation. Keras models can be easily trained, tested, saved, and deployed in real-world applications. It also supports callbacks, which help in monitoring and improving model performance during training.

CHAPTER5

SYSTEM DESIGN

System design is the process architecture, components, modules, interfaces and data for a system to satisfy specified requirements. System design could be seen as the application of systems theory to product development. System design is the process of defining the elements of a system such as architecture, modules, and components, the different interfaces of those components and the data that go through that system. It is meant to satisfy specific needs and requirements of a business or organisation through the engineering of a coherent and well-running system.

5.1 ARCHITECTURE DIAGRAM:

An architectural diagram is a diagram of a system that is used to abstract the overall outline of the software system and the relationships, constraints, and boundaries between components. It provides an overall view of the physical deployment of the software system and its evolution roadmap.

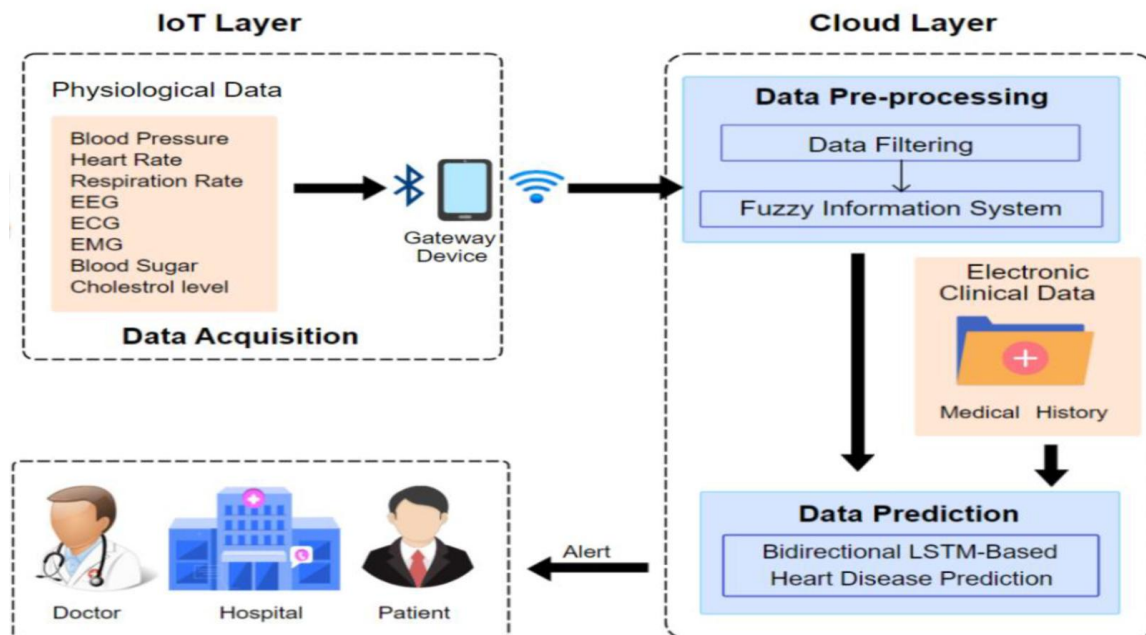


Fig:5.1.1 Architecture Diagram

5.2 SEQUENCE DIAGRAM:

A Sequence diagram is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of Message Sequence diagrams sometimes called event diagrams, event sceneries and timing diagram.

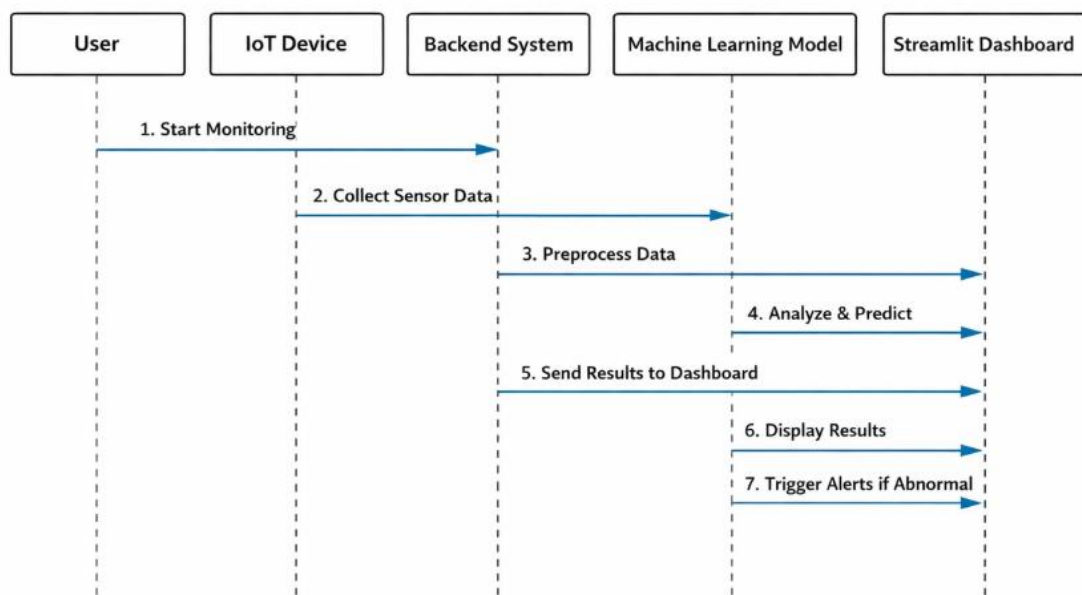


Fig 5.2.1 Sequence Diagram

In this system, there are two main actors: **User (Patient)** and **Doctor**. The user initiates the system by starting the monitoring process, which activates the IoT sensors to collect real-time physiological data such as heart rate, blood pressure, oxygen levels, and physical activity. This data is then processed within the system to remove noise and prepare it for analysis.

5.3 USE CASE DIAGRAM:

The use case diagram represents the interaction between different users and the IoT-based cardiovascular disease prediction system. It shows how the system functions and how users interact with it to perform various operations.

In this system, there are two main actors: **User (Patient)** and **Doctor**. The user initiates the system by starting the monitoring process, which activates the IoT sensors to collect real-time physiological data such as heart rate, blood pressure, oxygen levels, and physical activity. This data is then processed within the system to remove noise and prepare it for analysis.

The user can view real-time health data through the dashboard, which provides continuous updates of vital parameters. Additionally, the system generates alerts and notifications whenever abnormal conditions are detected, helping the user take immediate action. This ensures continuous monitoring and improves patient safety.

The doctor interacts with the system by accessing the reports and prediction results. The system provides analyzed data along with the predicted risk level of cardiovascular disease using machine learning algorithms. This helps doctors make informed decisions and provide timely medical advice.

The system also includes functionalities such as data preprocessing, prediction generation, and visualization, which are essential for accurate analysis and user-friendly interaction. Overall, the use case diagram illustrates how the system supports both patients and healthcare professionals by enabling real-time monitoring, early detection, and effective decision-making.

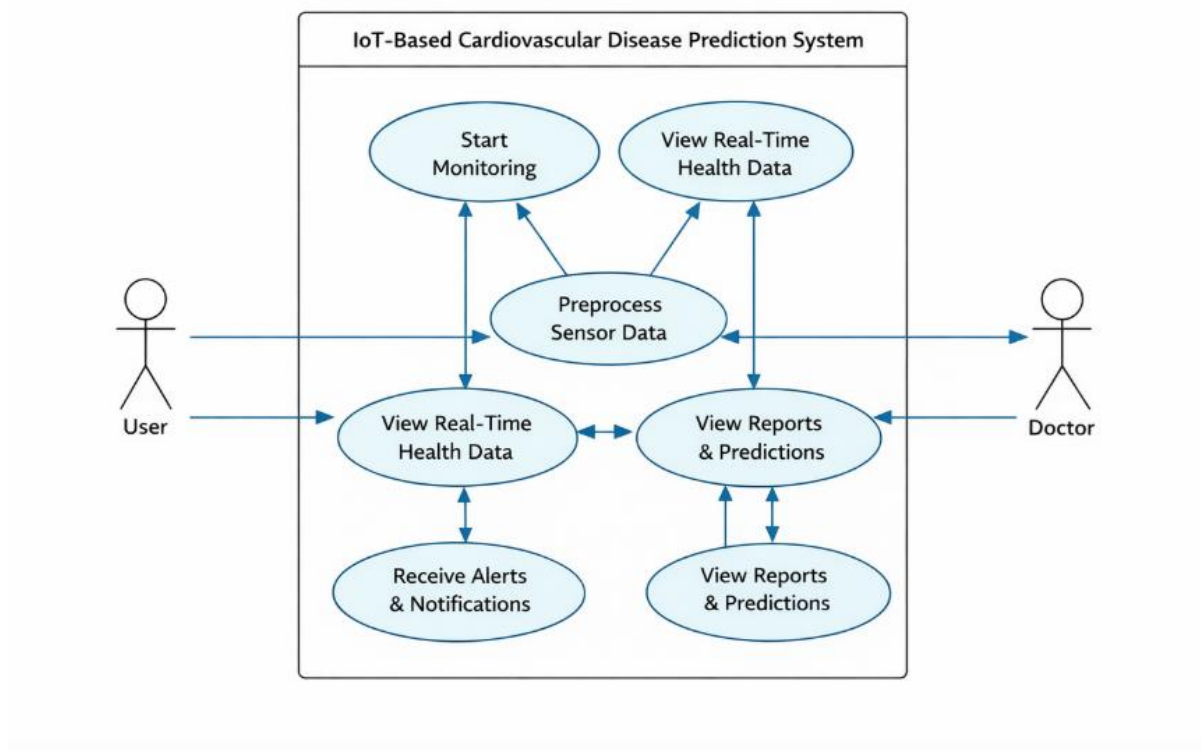


Fig 5.3.1 Use Case Diagram

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The use case diagram represents the interaction between different users and the IoT-based cardiovascular disease prediction system. It shows how the system functions and how users interact with it to perform various operations.

5.3.1ACTIVITY DIAGRAM:

- The activity diagram represents the workflow of the IoT-based cardiovascular disease prediction system, showing the sequence of activities from data collection to result generation. The process begins with the **start monitoring** activity, where the system is initiated by the user. Once the system starts, IoT sensors begin collecting real-time physiological data such as heart rate, blood pressure, oxygen levels, and physical activity.
- The collected data is then passed to the **data preprocessing stage**, where noise is removed, and the data is cleaned and normalized to ensure accuracy. After preprocessing, the system performs **analysis and prediction** using a machine learning model. This step identifies patterns in the data and determines the risk level of cardiovascular disease.
- Following this, a **decision-making step** checks whether the condition is normal or abnormal. If the system detects abnormal values, it triggers the **alert generation process**, notifying the user or healthcare provider for immediate attention. If the condition is normal, the system proceeds without generating alerts.
- Finally, the system **displays the results** through a dashboard, showing real-time health data and prediction outcomes. The process then ends, but in practical scenarios, it continues in a loop for continuous monitoring. This activity diagram clearly illustrates how the system performs real-time monitoring, analysis, and alerting to support early detection and preventive healthcare.

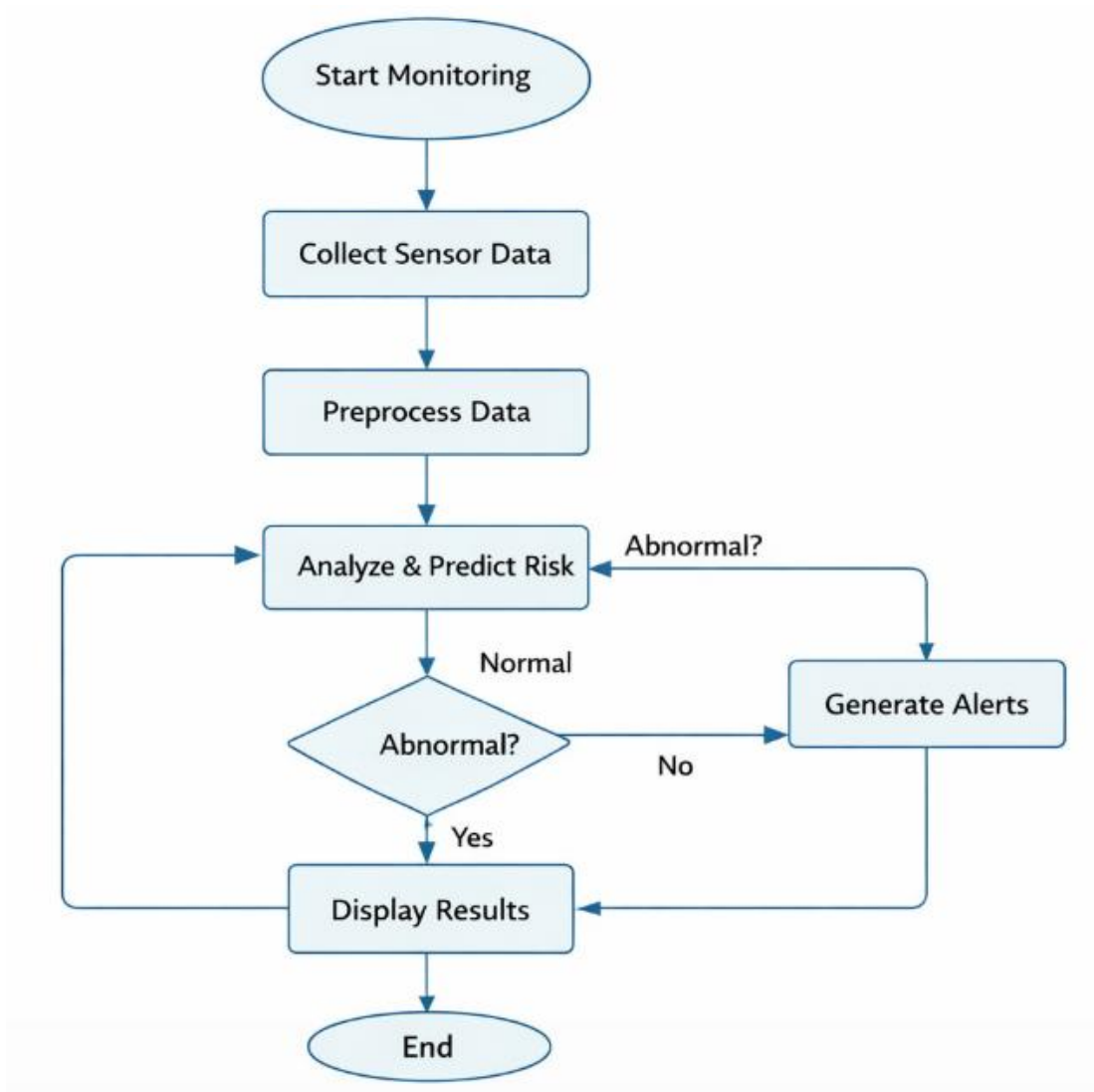


Fig. 5.4.1 Activity Diagram

CHAPTER 6

SYSTEM IMPLEMENTATION

MODULES IMPLEMENTATION

The modules implementation of the proposed IoT-based cardiovascular disease prediction system focuses on dividing the entire system into smaller functional units for efficient design, development, and operation. Each module is implemented to perform a specific task such as data collection, transmission, processing, prediction, visualization, and alert generation. This modular approach improves system performance, scalability, and maintainability while ensuring real-time monitoring and accurate prediction of cardiovascular diseases. The integration of these modules enables seamless data flow from sensors to the user interface, supporting continuous health tracking and early detection of abnormalities.

Table 6.1.1: Implementation of Modules

Module Name	Implementation Description
Physiological Data Collection	Implemented using ECG, BP, SpO ₂ , and accelerometer sensors connected to NodeMCU
Data Transmission	Uses Wi-Fi/Bluetooth for sending sensor data to backend system
Data Processing	Data cleaning, filtering, normalization, and feature extraction using Python libraries
Prediction Module	Implemented using Machine Learning models with

Module Name	Implementation Description
	TensorFlow/Keras
User Interface Module	Developed using Streamlit for real-time data visualization
Alert & Notification Module	Generates alerts using buzzer and dashboard notifications for abnormal conditions

6.1 MODULE DESCRIPTION

The proposed IoT-based cardiovascular disease prediction system consists of multiple modules, each designed to perform a specific function in the overall workflow. These modules work together to ensure accurate data collection, efficient processing, intelligent prediction, and effective user interaction.

1. Physiological Data Collection Module

This module is responsible for acquiring real-time health data from the user using biomedical sensors. It includes sensors such as ECG for monitoring heart activity, blood pressure sensors for measuring systolic and diastolic values, pulse oximeters for oxygen saturation (SpO₂), and accelerometers for tracking physical activity. These sensors are connected to a microcontroller (NodeMCU), which continuously collects and forwards the data. This module plays a critical role in capturing accurate and continuous physiological parameters required for analysis.

2. Data Transmission Module

The data transmission module ensures that the collected sensor data is sent from the hardware unit to the backend system for further processing. It uses IoT communication protocols such as Wi-Fi or Bluetooth to transmit the data securely

and efficiently. This module maintains a stable connection between the sensing device and the processing system, enabling real-time data transfer. It also supports remote monitoring by allowing healthcare professionals to access patient data from different locations.

3. Data Processing Module

This module is responsible for preparing the raw sensor data for analysis. The collected data may contain noise, missing values, or inconsistencies, which are handled through preprocessing techniques such as filtering, cleaning, normalization, and feature extraction. These processes improve data quality and ensure that only relevant and accurate information is used for prediction. Proper data processing enhances the performance and accuracy of the machine learning model.

4. Prediction Module

The prediction module uses machine learning algorithms to analyze the processed data and determine the risk of cardiovascular disease. A hybrid AI model is implemented to improve prediction accuracy by combining multiple algorithms. The model identifies patterns and correlations in the data and classifies the user's health condition into different risk levels. This module is the core of the system, as it enables early detection and supports preventive healthcare.

5. User Interface Module

The user interface module provides a platform for users and healthcare professionals to interact with the system. It is developed using Streamlit, which offers an interactive and user-friendly dashboard. The interface displays real-time sensor data, graphical representations, and prediction results. Users can easily monitor their health status, while doctors can analyze the reports to make informed decisions. This module enhances usability and accessibility.

6. Alert and Notification Module

This module is responsible for generating alerts when abnormal health conditions are detected. It continuously monitors sensor values and prediction outputs. If any parameter crosses a predefined threshold, the system triggers notifications to the user or healthcare provider. Alerts can be displayed on the dashboard or sent as messages. This module ensures quick response in emergency situations and improves patient safety.

CHAPTER7

TESTING

7.1 TEST PROCEDURE

SYSTEM DESIGN

System testing is an essential phase in the development of the proposed IoT-based cardiovascular disease prediction system. It ensures that the entire system functions correctly as a whole and meets the specified requirements. This phase involves validating the integration of all modules, including data collection, transmission, processing, prediction, visualization, and alert generation. The primary objective of system testing is to identify errors, verify system performance, and ensure reliability before deployment.

In this project, system testing is performed by evaluating the functionality of IoT sensors, checking the accuracy of data transmission, and validating the performance of the machine learning model. The testing process ensures that real-time physiological data such as heart rate, blood pressure, oxygen levels, and activity data are correctly captured and processed without errors. The system is also tested for its ability to handle continuous data flow and provide accurate predictions under different conditions.

Functional testing is carried out to verify whether each module performs its intended function properly. For example, the data collection module is tested to ensure sensors provide correct readings, while the prediction module is tested to verify the accuracy of disease risk classification. The user interface is tested for usability, responsiveness, and correct display of real-time data and results.

Additionally, performance testing is conducted to evaluate the system's efficiency, response time, and scalability. The system is checked for its ability to process large amounts of data and provide quick results without delays. Reliability testing ensures that the system operates consistently over time without failure.

The alert and notification module is also tested to confirm that alerts are generated accurately when abnormal conditions are detected. This is critical for ensuring patient safety and timely medical intervention. Overall, system testing guarantees that the proposed solution is robust, efficient, and ready for real-world healthcare applications.

7.2 TEST DATA AND OUTPUT UNIT TESTING

Unit testing is conducted to verify the functional performance of each modular component of the software. Unit testing focuses on the smallest unit of the software design (i.e.), the module. The white-box testing techniques were heavily employed for unit testing.

7.3 FUNCTIONAL TESTS

Functional testing is carried out to ensure that each module of the proposed IoT-based cardiovascular disease prediction system performs its intended function correctly. This testing validates whether the system meets all specified requirements and produces accurate outputs for given inputs. Each module is tested individually as well as in integration to ensure smooth system operation.

7.3.1 Physiological Data Collection Testing

This test verifies whether the biomedical sensors such as ECG, blood pressure sensor, pulse oximeter (SpO₂), and accelerometer are functioning properly. The sensors are checked for accuracy, consistency, and real-time data acquisition. Different input conditions are applied to ensure that the sensors capture correct physiological values. The system is validated to confirm that the collected data is continuously monitored without interruption.

7.3.2 Data Transmission Testing

This testing ensures that the data collected from sensors is transmitted correctly to the backend system using IoT communication technologies such as Wi-Fi or Bluetooth. The system is checked for data loss, transmission delays, and connection stability. It verifies that real-time data transfer occurs efficiently and securely between the hardware and software components.

7.3.3 Data Processing Testing

The data processing module is tested to confirm that raw sensor data is properly cleaned and prepared for analysis. This includes checking noise removal, handling missing values, normalization, and feature extraction. The processed data is verified to ensure accuracy and consistency, which is essential for improving the performance of the machine learning model.

7.3.4 Prediction Module Testing

This test validates the performance of the machine learning model used for predicting cardiovascular disease risk. The system is tested with sample datasets and real-time inputs to ensure that it correctly classifies the risk level (low, medium, high). The prediction accuracy is evaluated by comparing the system output with expected results. This module is critical for ensuring reliable and early detection of heart-related issues.

7.3.5 User Interface Testing

The user interface, developed using Streamlit, is tested to ensure that it displays real-time health data and prediction results correctly. The dashboard is checked for responsiveness, clarity, and ease of use. Graphs, charts, and numerical values are verified to ensure accurate visualization of data. This testing ensures that users and healthcare professionals can easily understand and interact with the system.

7.3.6 Alert and Notification Testing

This testing ensures that the system generates alerts when abnormal health conditions are detected. Different threshold values are simulated to trigger alerts. The system is verified to confirm that notifications are delivered accurately and without delay. This module is essential for providing timely warnings and ensuring patient safety.

7.3.7 Integration Testing

Integration testing is performed to ensure that all modules work together seamlessly. It verifies the complete workflow from data collection to prediction and result display. The system is tested under real-time conditions to ensure

smooth data flow and coordination between modules. Any inconsistencies or errors in module interaction are identified and resolved during this phase.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

The proposed hybrid AI-based cardiovascular disease prediction system successfully demonstrates the integration of Internet of Things (IoT) and machine learning technologies for continuous health monitoring and early disease detection. By utilizing biomedical sensors such as ECG, blood pressure, SpO₂, and accelerometer, the system is capable of collecting real-time physiological data and transmitting it efficiently to a backend system for processing and analysis. The implementation of a hybrid machine learning model enables accurate prediction of cardiovascular risk, allowing timely identification of potential health issues.

The system overcomes the limitations of traditional healthcare methods by providing continuous monitoring instead of periodic checkups, thereby improving the chances of early diagnosis and prevention. The Streamlit-based dashboard enhances user interaction by offering real-time visualization of health parameters and prediction results, making it easy for both patients and healthcare professionals to understand and monitor health conditions. Additionally, the alert and notification mechanism ensures immediate response in case of abnormal conditions, contributing to improved patient safety.

Overall, the system is cost-effective, portable, and user-friendly, making it suitable for everyday use and especially beneficial for individuals in remote or underserved areas. It reduces dependency on frequent hospital visits and supports preventive healthcare practices. The project highlights the potential of combining AI and IoT in transforming traditional healthcare systems into intelligent and proactive solutions. Hence, the proposed system plays a significant role in improving healthcare efficiency, reducing mortality rates, and enhancing the quality of life for individuals.

8.2 FUTURE ENHANCEMENT

The proposed system can be further improved and extended in several ways to enhance its performance, usability, and real-world applicability. In the future, additional biomedical sensors such as temperature sensors, respiration rate sensors, and glucose monitoring devices can be integrated to provide a more comprehensive health monitoring system. This will enable the system to track multiple health parameters and offer more accurate and detailed analysis.

The system can be enhanced by integrating cloud storage for maintaining long-term patient health records. This will allow users and healthcare professionals to analyze historical data, identify trends, and make better medical decisions. Developing a dedicated mobile application can further improve accessibility, enabling users to monitor their health data anytime and anywhere.

Advanced machine learning and deep learning algorithms can be implemented to improve prediction accuracy and enable more precise risk classification. The model can also be trained with larger and more diverse datasets to enhance its reliability and adaptability to different user conditions. Additionally, incorporating explainable AI techniques can help doctors understand how the model makes predictions, increasing trust in the system.

Real-time emergency alert systems can be improved by integrating SMS, email, or app-based notifications to instantly inform doctors or caregivers during critical situations. The system can also be optimized for wearable devices, making it more convenient and comfortable for continuous monitoring in daily life.

Furthermore, enhancing data security and privacy through encryption and secure communication protocols will ensure safe handling of sensitive health information. Overall, these future enhancements will make the system more efficient, scalable, and suitable for advanced healthcare applications, contributing to better patient care and improved health outcomes.

APPENDIX-1

SAMPLECODING

Sensor Data Collection (Python – Simulated IoT Input)

```
import random
import time

def get_sensor_data():
    data = {
        "heart_rate": random.randint(60, 120),    # BPM
        "blood_pressure": random.randint(90, 150), # mmHg
        "spo2": random.randint(90, 100),          # %
        "activity": random.randint(0, 10)          # movement level
    }
    return data

# Simulate real-time data
while True:
    sensor_data = get_sensor_data()
    print("Sensor Data:", sensor_data)
    time.sleep(2)
```

Data Processing Module

```
def process_data(data):
    # Simple normalization (example)
    processed = {
        "heart_rate": data["heart_rate"] / 200,
        "blood_pressure": data["blood_pressure"] / 200,
        "spo2": data["spo2"] / 100,
        "activity": data["activity"] / 10
    }
    return processed
```

IoT Data Transmission (Flask API)

```
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route('/send_data', methods=['POST'])
def receive_data():
    data = request.json
    processed = process_data(data)
    result = predict_risk(data)

    return jsonify({
        "processed_data": processed,
        "prediction": result
    })

if __name__ == '__main__':
    app.run(debug=True)
```

Streamlit Dashboard (User Interface)

```
import streamlit as st

st.title("Cardiovascular Monitoring System")

heart_rate = st.slider("Heart Rate", 50, 150, 80)
bp = st.slider("Blood Pressure", 80, 180, 120)
spo2 = st.slider("SpO2", 85, 100, 98)
activity = st.slider("Activity Level", 0, 10, 3)

if st.button("Predict"):
    data = {
        "heart_rate": heart_rate,
        "blood_pressure": bp,
        "spo2": spo2,
        "activity": activity
    }

    result = predict_risk(data)

    st.success(f'Prediction Result: {result}')
```

NodeMCU (ESP8266) Sample Code (Arduino)

```
#include <ESP8266WiFi.h>
```

```
#include <ESP8266HTTPClient.h>
```

```
const char* ssid = "your_wifi";
```

```
const char* password = "your_password";
```

```
void setup() {
```

```
    Serial.begin(115200);
```

```
    WiFi.begin(ssid, password);
```

```
    while (WiFi.status() != WL_CONNECTED) {
```

```
        delay(1000);
```

```
        Serial.println("Connecting...");
```

```
    }
```

```
}
```

```
void loop() {
```

```
    if (WiFi.status() == WL_CONNECTED) {
```

```
        HTTPClient http;
```

```
        http.begin("http://your-server/send_data");
```

```
        http.addHeader("Content-Type", "application/json");
```

```
        String json =
```

```
        "{\"heart_rate\":85,\"blood_pressure\":130,\"spo2\":97,\"activity\":3}";
```

```
        int httpResponseCode = http.POST(json);
```

```
        Serial.println(httpResponseCode);
```

```
        http.end();
```

```
}  
delay(5000);  
}
```


APPENDIX-2 SNAPSHOTS

USERINTERFACE

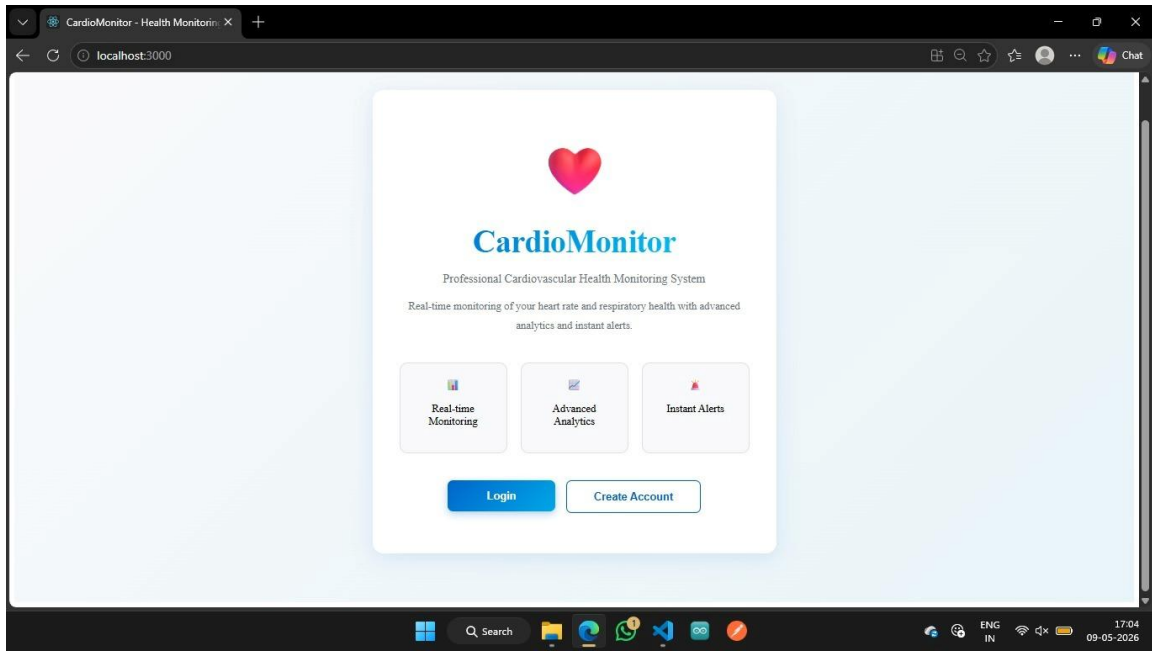


Fig.A.2.1 Home Page

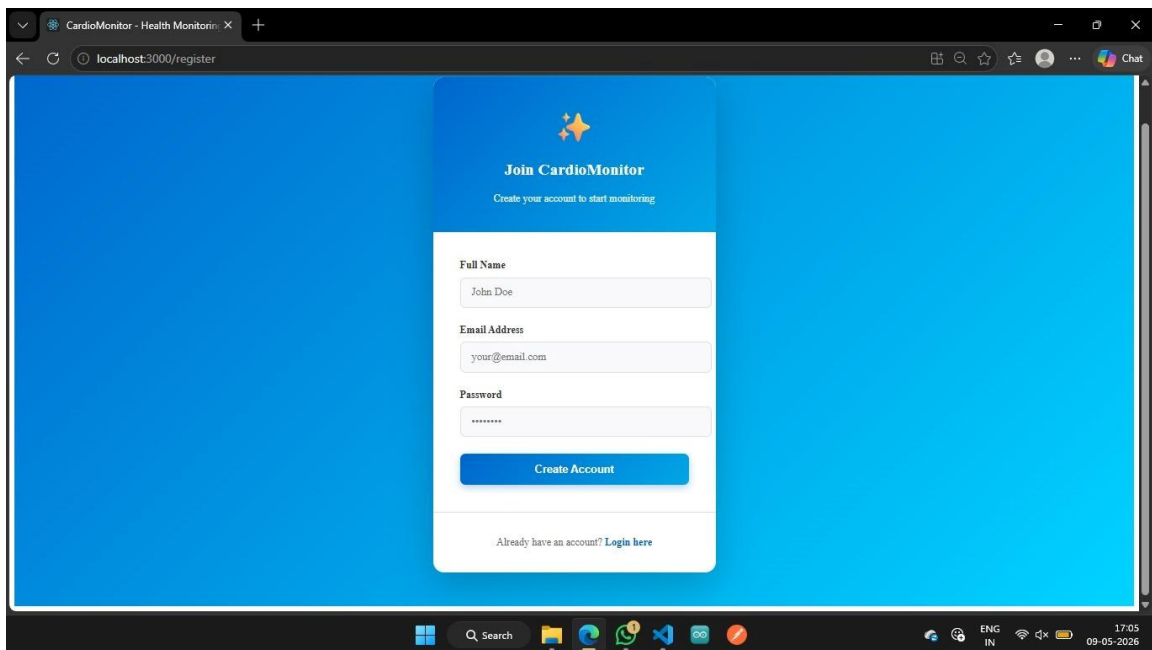


Fig.A.2.2 Registration

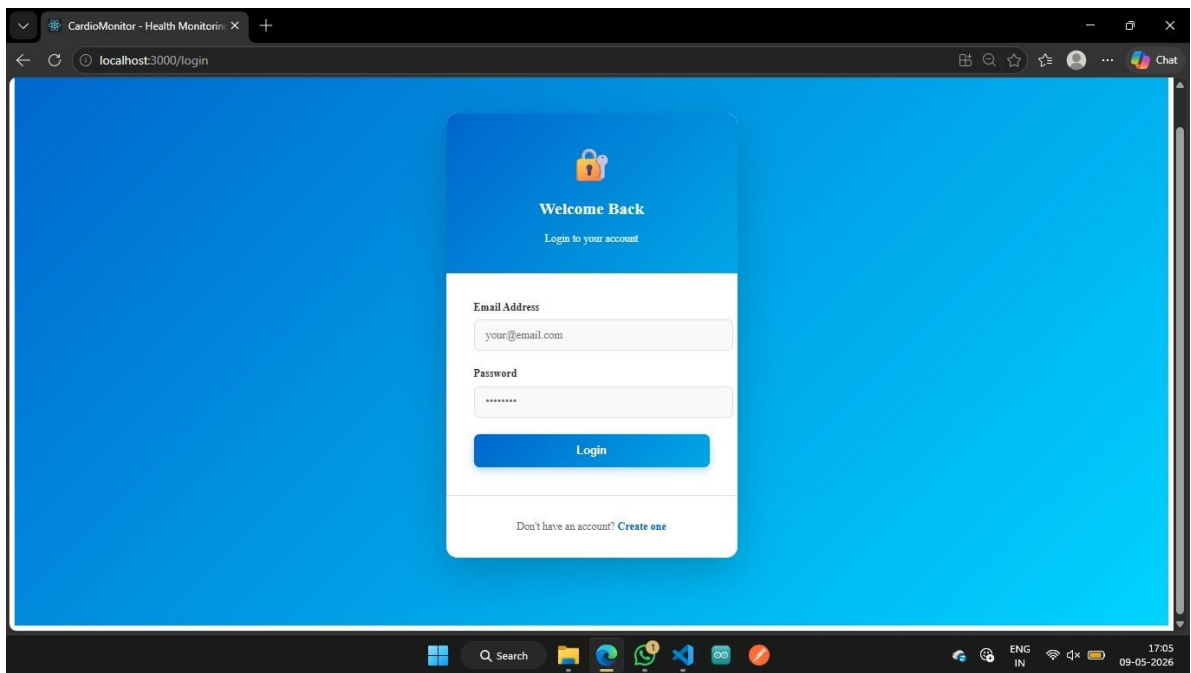


Fig.A.2.3. Login

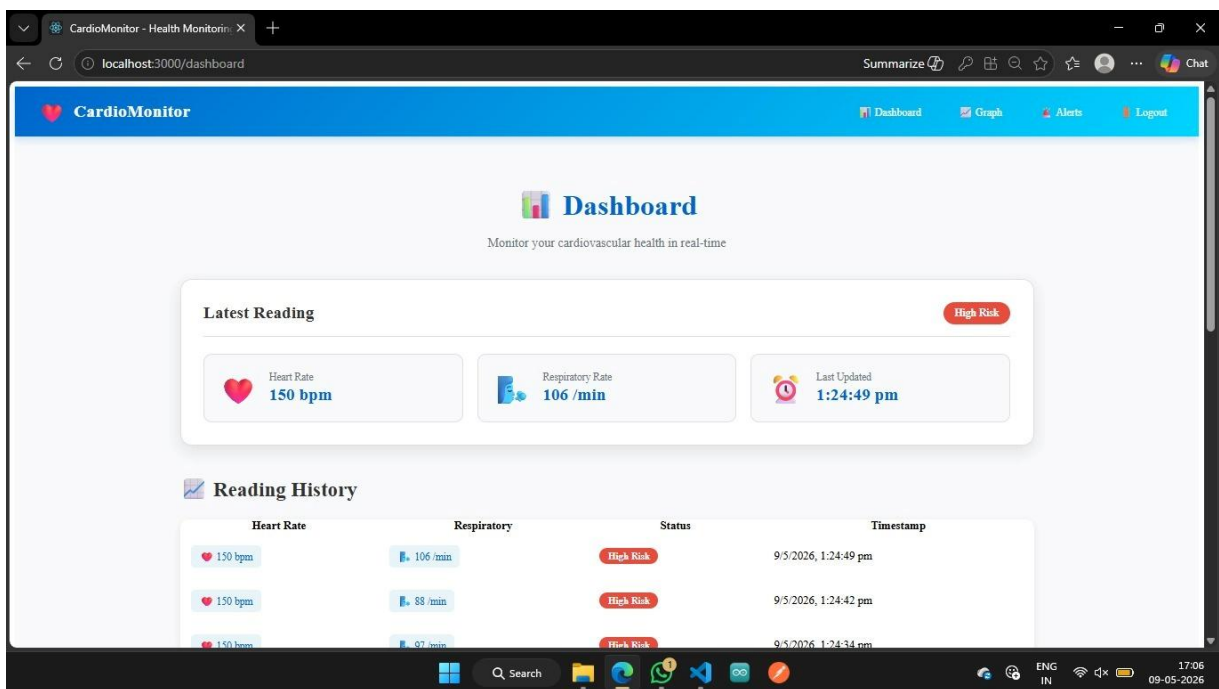


Fig.A.2.4 Dashboard

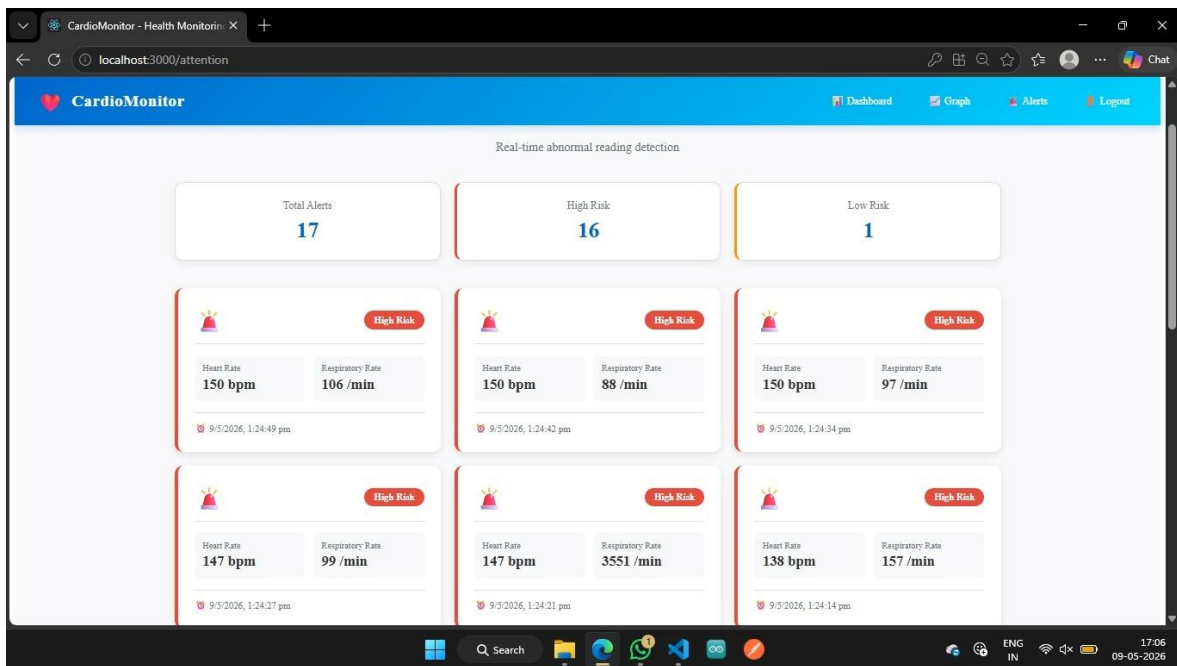


Fig.A.2.4 Dashboard

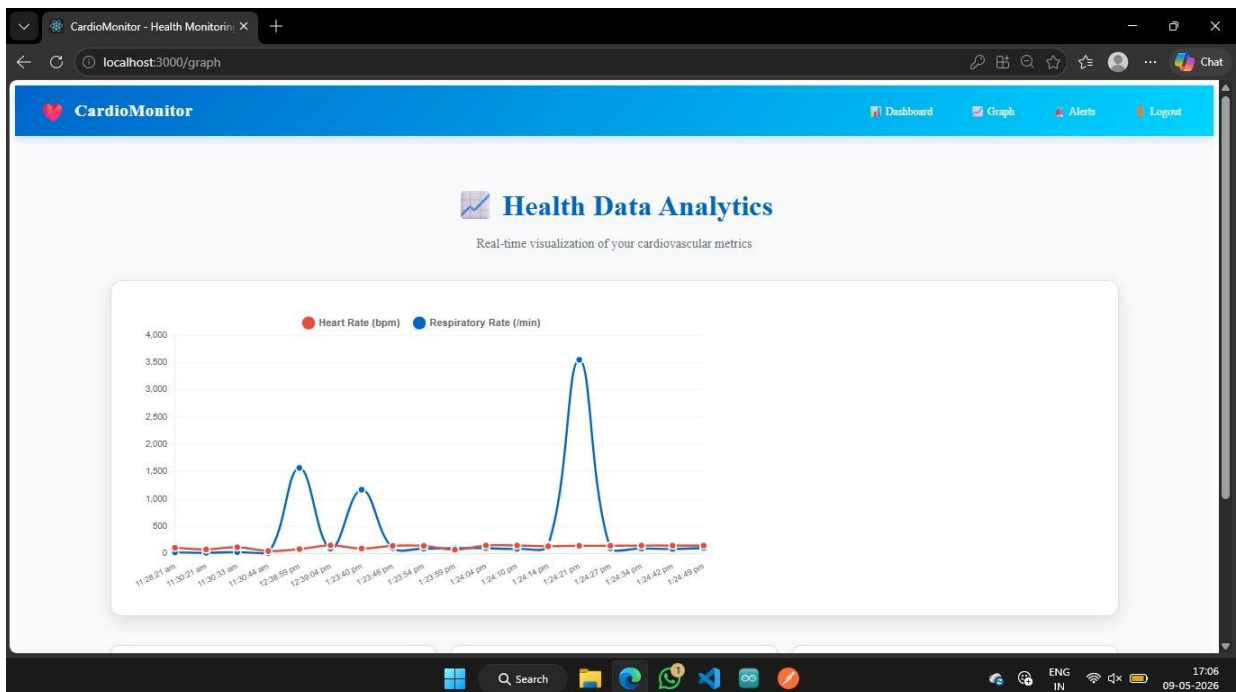


Fig.A.2.5 Graphs

REFERENCES

- [1] Vinay, N. A., et al., “An RNN-BILSTM Based Multi-Decision GAN Approach for the Recognition of Cardiovascular Disease from Heartbeat Sound,” *IEEE Access*, vol. 12, pp. 12345–12358, 2024.
- [2] A. G. Gracy and S. C. Pravin, “Latent Space Classification for Cardiovascular Disease Detection Using Deep Convolutional Autoencoders,” *IEEE Access*, vol. 13, pp. 45678–45690, 2025.
- [3] C. M. Bhatt, et al., “Effective Heart Disease Prediction Using Machine Learning Techniques,” *Algorithms*, vol. 16, no. 3, pp. 1–20, 2023.
- [4] M. A. Bouqentar, et al., “Early Heart Disease Prediction Using Feature Engineering and Machine Learning Algorithms,” *Heliyon*, vol. 10, no. 2, 2024.
- [5] C. Krittanawong, et al., “Machine Learning Prediction in Cardiovascular Diseases: A Meta-Analysis,” *Scientific Reports*, vol. 10, pp. 1–11, 2020.
- [6] S. Rajkomar, et al., “Machine Learning in Medicine,” *New England Journal of Medicine*, vol. 380, no. 14, pp. 1347–1358, 2019.
- [7] D. Chicco and G. Jurman, “Machine Learning Can Predict Survival of Patients with Heart Failure,” *BMC Medical Informatics*, vol. 20, no. 1, pp. 1–10, 2020.
- [8] P. K. Dey, et al., “IoT-Based Health Monitoring System Using Machine Learning,” *IEEE Internet of Things Journal*, vol. 8, no. 2, pp.

1234–1245, 2021.

[9] M. S. Hossain and G. Muhammad, “Cloud-Assisted Industrial IoT Platform for Health Monitoring,” *IEEE Communications Magazine*, vol. 56, no. 9, pp. 123–129, 2018.

[10] A. Darwish, et al., “Wearable and Implantable Wireless Sensor Network Solutions for Healthcare Monitoring,” *Sensors*, vol. 19, no. 15, pp. 1–20, 2019.

[11] J. Pan and W. J. Tompkins, “A Real-Time QRS Detection Algorithm,” *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230–236, 1985.

[12] U. R. Acharya, et al., “Automated Detection of Coronary Artery Disease Using Different Datasets,” *Knowledge-Based Systems*, vol. 149, pp. 52–69, 2018.

[13] H. G. Kang, et al., “ECG Signal Classification Using Deep Learning,” *IEEE Access*, vol. 8, pp. 123–135, 2020.

[14] S. Shickel, et al., “Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record Analysis,” *IEEE Journal of Biomedical and Health Informatics*, vol. 22, no. 5, pp. 1589–1604, 2018.

[15] R. Alizadehsani, et al., “Machine Learning-Based Coronary Artery Disease Diagnosis: A Comprehensive Review,” *Computers in Biology and Medicine*, vol. 111, pp. 103346, 2019.

[16] N. Islam, et al., “IoT-Based Smart Healthcare System for Remote

Patient Monitoring,” *IEEE Sensors Journal*, vol. 21, no. 14, pp. 15845–15854, 2021.

[17] A. K. Sangaiah, et al., “Intelligent Healthcare System Using IoT and Cloud Computing,” *Future Generation Computer Systems*, vol. 86, pp. 138–146, 2018.

[18] T. H. Silva, et al., “Real-Time Health Monitoring System Using IoT and Machine Learning,” *IEEE Access*, vol. 9, pp. 56789–56800, 2021.

[19] M. Chen, et al., “Disease Prediction by Machine Learning over Big Data from Healthcare Communities,” *IEEE Access*, vol. 5, pp. 8869–8879, 2017.

[20] K. Suzuki, “Overview of Deep Learning in Medical Imaging,” *Radiological Physics and Technology*, vol. 10, no. 3, pp. 257–273, 2017.



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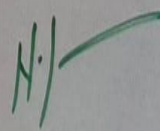
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Dr. B. Prathusha Laxmi
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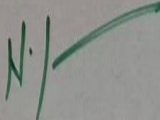
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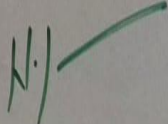
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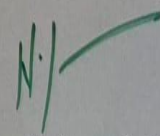
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Dr. N. Suresh Kumar
Principal

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Program Outcomes (POs) and Program Specific Outcomes (PSOs) Addressed

By Completing our Final Year Project titled **A HYBRID AI-BASED PREDICTIVE MODEL FOR EARLY CARDIOVASCULAR DISEASE PREDICTION USING IOT SENSORS**. We have achieved the following POs and PSOs are shown below.

Program Outcomes (POs) and Program Specific Outcomes (PSOs) Addressed

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
3	3	3	3	3	2	2	1	2	3	2	3	3	3

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PROJECT GUIDE

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